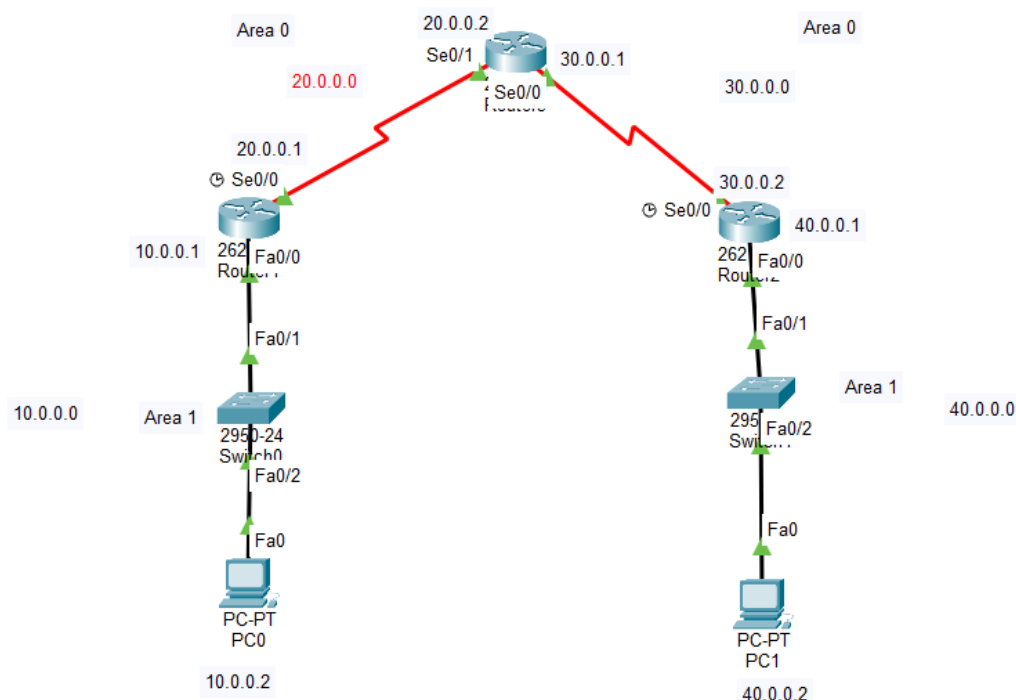


Day 5: Configuring OSPF Practical: Configure OSPF on a network with multiple areas. Task: Simulate a scenario where a router fails and observe OSPF reconvergence.

Task 1.

Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.



Task 2.

Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command.

Router 1

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:36	20.0.0.1	Serial0/1
1.1.1.2	0	FULL/ -	00:00:31	30.0.0.2	Serial0/0

```
Router#
```

Router 2

```
Router#  
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
30.0.0.1	0	FULL/ -	00:00:39	20.0.0.2	Serial0/0

```
Router#
```

Router 3

```
Router>  
Router>en  
Router#show ip ospf ne  
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
30.0.0.1	0	FULL/ -	00:00:30	30.0.0.1	Serial0/0

```
Router#  
Router#
```

Task 3.

Add two PCs to different areas in your OSPF network, assign them IP addresses, and test connectivity by pinging from one PC to the other.

```
C:\>ping 40.0.0.2  
  
Pinging 40.0.0.2 with 32 bytes of data:  
  
Reply from 40.0.0.2: bytes=32 time=55ms TTL=125  
Reply from 40.0.0.2: bytes=32 time=60ms TTL=125  
Reply from 40.0.0.2: bytes=32 time=48ms TTL=125  
Reply from 40.0.0.2: bytes=32 time=64ms TTL=125  
  
Ping statistics for 40.0.0.2:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 48ms, Maximum = 64ms, Average = 56ms
```

Task 4.

Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf' commands to observe how OSPF reconverges and restores connectivity after re-enabling the interface.

Hint: Track the time it takes for OSPF to update the routing tables after the failure and recovery.

```
C:\>ping 40.0.0.2

Pinging 40.0.0.2 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.

Ping statistics for 40.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 40.0.0.2

Pinging 40.0.0.2 with 32 bytes of data:

Reply from 40.0.0.2: bytes=32 time=45ms TTL=125
Reply from 40.0.0.2: bytes=32 time=48ms TTL=125
Reply from 40.0.0.2: bytes=32 time=45ms TTL=125
Reply from 40.0.0.2: bytes=32 time=39ms TTL=125

Ping statistics for 40.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 39ms, Maximum = 48ms, Average = 44ms
```

With Failure

```
Router(config)#int serial 0/1
Router(config-if)#shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to down

01:08:13: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/1 from FULL to DOWN, Neighbor Down: Interface or detached

Router(config-if)#ex
Router(config)#show ip route
^
% Invalid input detected at '^' marker.

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    30.0.0.0/8 is directly connected, Serial0/0
O IA 40.0.0.0/8 [110/65] via 30.0.0.2, 00:29:14, Serial0/0
```

```

Router#show ip ospf
Routing Process "ospf 1" with ID 30.0.0.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 1
    Area has no authentication
    SPF algorithm executed 7 times
    Area ranges are
    Number of LSA 5. Checksum Sum 0x024c3e
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
Router#

```

Without Failure

```

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O IA 10.0.0.0/8 [110/65] via 20.0.0.1, 00:00:10, Serial0/1
C   20.0.0.0/8 is directly connected, Serial0/1
C   30.0.0.0/8 is directly connected, Serial0/0
O IA 40.0.0.0/8 [110/65] via 30.0.0.2, 00:34:07, Serial0/0

Router#show ip ospf
Routing Process "ospf 1" with ID 30.0.0.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 2
    Area has no authentication
    SPF algorithm executed 9 times
    Area ranges are
    Number of LSA 5. Checksum Sum 0x024176
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
Router#

```

With Shutdown

Event	Time to record
Failure detection time	1 second
Neighbor re-establishment	5–10 seconds
Route restoration	5–15 seconds
Connectivity restoration	5–15 seconds

No Shutdown(re-enabling)

Event	Time
Failure detection time	1 second
Neighbor re-establishment	323 seconds (5 min 23 sec)
Route restoration	5–15 seconds
Connectivity restoration	5–15 seconds